Supply Chain & Delivery Performance Analysis

1. Project Overview

This project analyzes supply chain operations using transactional and logistics data. The goal is to identify inefficiencies in inventory management, delivery performance, and transportation cost to support better business decisions.

2. Dataset Summary

- **Rows: (your dataset count)**
- **Columns: 15**
- **Key Features:**
    - **Order details (Date, Region, Category, Supplier)**
    - **Inventory metrics (Inventory Level, Warehouse Capacity)**
    - **Logistics data (Transportation Cost, Lead Time)**
    - **Performance indicators (Backorder, Order Accuracy)**

3. Data Processing using Python

- **Loaded dataset using pandas**
- **Checked structure using df.info() and summary statistics**
- **Converted date column into proper format**
- **Verified data consistency**
- **Prepared cleaned dataset for SQL analysis**

4. Data Analysis using SQL

1. Regional Sales Performance

Analyzed total units sold across different regions.

Region	total_units_sold
North	170175
West	168452
East	153047
South	148997

2. Category-wise Inventory Analysis

Evaluated inventory distribution across product categories.

Category	total_revenue
Electronics	19207238.505421996
Clothing	18789437.139704008
Furniture	17227460.130157
Accessories	16020440.245646004

3. Transportation Cost Analysis

Analyzed transportation cost across regions and categories.

Supplier	total_transport_cost
Supplier B	2391250.1584276003
Supplier A	2382512.8571847007
Supplier D	2380107.3411675994
Supplier C	2114351.8357747006

4. Delivery Performance

Calculated average lead time to evaluate delivery efficiency.

Region	avg_lead_time
North	16.11
West	15.78
South	15.53
East	15.49

5. Backorder Analysis

Identified frequency of backorders in the system.

delay_status	total_orders
Delayed	1016
On Time	184

6. Warehouse Utilization

Measured how efficiently warehouse capacity is used.

Backorder	total_orders
FALSE	1084
TRUE	116

7. Inventory Distribution by Region

Analyzed how inventory is distributed across regions.

Warehouse	utilization_percent
Warehouse 2	36.66
Warehouse 1	35.02
Warehouse 3	34.80

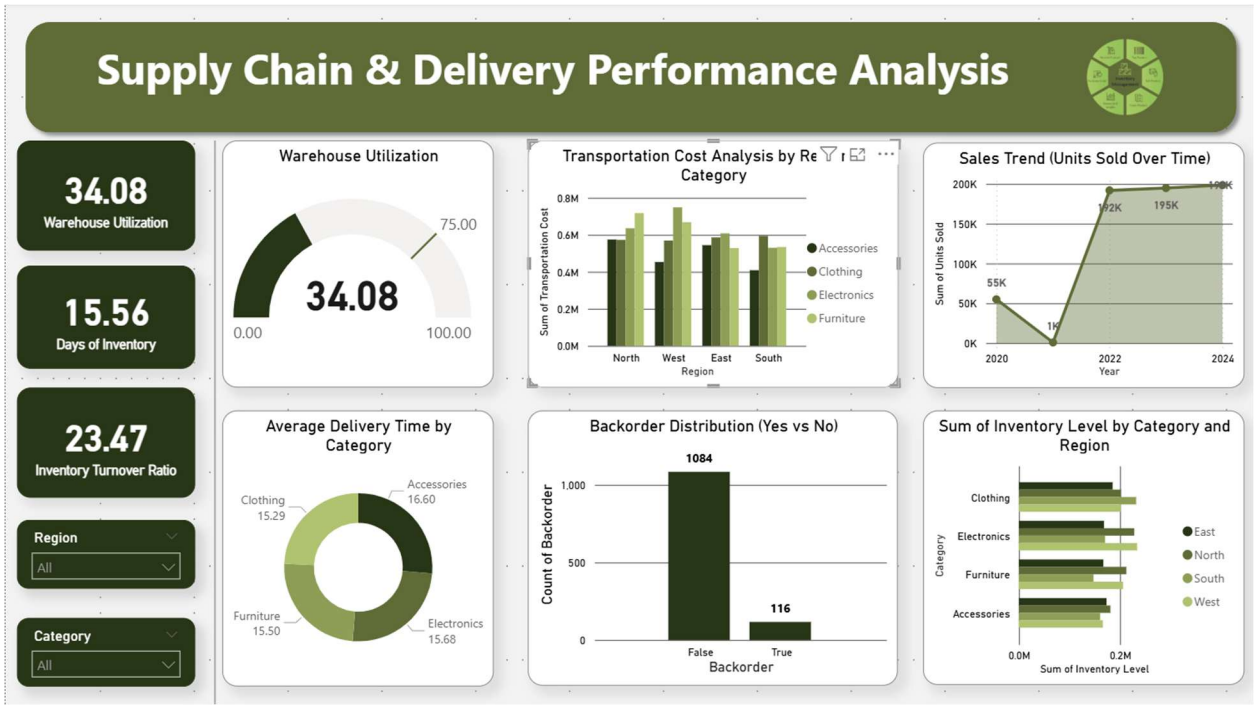
8. Sales Trend Over Time

Analyzed units sold across years to identify trends.

Region	total_inventory
North	823947
West	807273
South	711022
East	694326

5. Dashboard in Power BI

An interactive dashboard was created in Power BI to visualize key supply chain metrics.



6. Business Recommendations

- **Improve delivery performance by optimizing routes in high lead-time regions**
- **Reduce transportation cost by reviewing supplier contracts and logistics planning**
- **Minimize backorders by improving demand forecasting and inventory replenishment**
- **Balance warehouse utilization by redistributing inventory across locations**
- **Focus on high-performing regions and categories for better revenue generation**